Model Solution and Computation

Liquidity Constraints in the U.S. Housing Market

Boar, Gorea, and Midrigan (ReStud, 2022)

Seoyeon Lim

The Ohio State University
lim.1523@osu.edu · Website

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The Computation in One View

- Fix prices and model parameters, then construct numerical state grids.
- Solve backward over age: calculate values and decisions for every grid state.
- Simulate forward: follow households as shocks and portfolio positions change.
- Use the simulated population to calculate moments and conduct experiments.

$$\boxed{\text{Finite life cycle} + \text{household risk} + \text{given prices}}$$

The baseline solution requires no aggregate-price or forecasting-rule iteration.

1 Numerical Representation

2 Backward Solution

3 Forward Simulation

Economic States and Computational States

At the baseline house price $p = 1$, mortgage states satisfy

$$\boxed{b = \omega\theta h, \qquad \bar{m} = \kappa_m \theta h, \qquad \kappa_m = \frac{r_m}{1 - (1 + r_m)^{-D}}.}$$

Code variable	Meaning
ω	Fraction of the original loan balance still outstanding
θ	Original financed balance relative to house value
h	House size, also house value at $p = 1$
$\widetilde{a} = (1 + r_l(a))a - \delta h$	Liquid resources after returns and maintenance

- The homeowner decision grid uses $(\widetilde{a}, \omega, \theta, h, z, e)$ at each age.
- A new mortgage resets $\omega' = 1$. Repayment lowers ω'.

State Grids in the Baseline Code

State or intermediate variable	Nodes	Range or construction
Liquid assets a	75	$[-0.4, 100]$, quadratic spacing
Adjusted liquid resources $\widetilde{a}$	75	Bounds implied by assets and maintenance
Liquidity after portfolio choice L	75	$[-1, 125]$, power spacing
Fraction outstanding ω	11	$[0, 1]$, uniform
Original mortgage ratio θ	5	$[0.25, 0.85]$, uniform
House size h	7	$[5, 40]$, power spacing
Persistent income z	9	Rouwenhorst approximation
Transitory income e	3	Gaussian quadrature

The model has $T = 244$ quarterly decision dates.
Nonuniform grids place more nodes at low assets and liquidity.

Approximating Household Risk

Income at age t is

$$y_t = \lambda_t z_t e_t, \qquad \log z_{t+1} = \rho_z \log z_t + \sigma_z \varepsilon_{t+1}.$$

- **Persistent income:** `rouwenhorst` returns 9 states and transition matrix F_{zz}.
- **Transitory income:** `qnwnorm` returns 3 log-income nodes and weights w_e.
- **Maintenance:** $\delta \in \{0, 0.0625\}$ with probabilities $(0.975, 0.025)$.
- **Mortgage utility cost:** an exponential draw, integrated analytically in `solveh.m`.

In this baseline, $\rho_z = 0.9908$ and $\sigma_z = 0.0761$.
The factor 0.55 in the transitory-variance adjustment is a measurement-error share, not persistent-income autocorrelation.

Three Value Objects Organize the Solution

Code arrays	Economic object	Role in the computation
`vbarh`, `vbarr`	Expected values	Integrate shocks for use as continuation values
`wh`, `wr`	Conditional values W_t	Optimize consumption and saving given the portfolio position and liquidity L
`vh`, `vr`	Decision-stage values V_t	Compare housing actions and integrate the current mortgage utility cost

Superscripts H and R distinguish homeowners and renters.
`ch` and `cr` store the associated conditional consumption policies.

① Numerical Representation

② Backward Solution

③ Forward Simulation

The Terminal Condition Starts the Recursion

At $T+1$, remaining resources enter the bequest payoff. For an owner,

$$w^{\text{estate}} = (1 + r_l(a))a + (1 - f_s)h - (1 + r_m)\omega\theta h - \delta h.$$

The code initializes

$$\bar{V}^H_{T+1} = \sum_{\delta} \pi_\delta \, B \frac{(\bar{w} + w^{\text{estate}})^{1-\sigma}}{1-\sigma}.$$

- Estate wealth includes housing sale proceeds net of mortgage repayment and maintenance.
- Renters leave liquid resources only.
- With the terminal payoff known, solve ages $T, T-1, \ldots, 1$ once each.

The Loop at Each Age

```
for t = p.T : -1 : 1
    % 1. Interpolate expected values at age t+1
    % 2. Optimize consumption; construct wh, wr
    % 3. Compare housing actions with solveh
    % 4. Integrate maintenance and income shocks
    % 5. Store vbarh(:,t), vbarr(:,t)
end
```

- The loop calculates decisions across the full numerical state space at each age.
- Age $t+1$ values summarize all later opportunities and risks.
- Each age uses the already solved next-age values. There is no value-function convergence loop over age t.

Conditional Consumption and Saving

Fix a homeowner's next mortgage and housing position. Let L denote liquidity remaining after that portfolio action. Then

$$W_t^H(L, \omega', \theta', h', z) = \max_c \left\{ u(c, h', n(c)) + \beta \bar{V}_{t+1}^H(a'(c), \omega', \theta', h', z) \right\}.$$

The home-production condition and budget imply

$$n(c) = \phi^{1/\gamma} c^{-\sigma/\gamma}, \qquad \boxed{a'(c) = L + \phi^{1+1/\gamma} c^{-\sigma/\gamma} - c.}$$

- `savings.m` computes a' for each candidate consumption choice.
- `wfunc.m` adds current utility to discounted continuation value.
- For renters, optimal rental spending also enters the budget.

▸ Renter problem

Bisection Bounds and Golden-Section Search

- First find feasible consumption bounds by solving $a'(c) = a_{\max}$ and $a'(c) = a_{\min}$ with bisection.
- Given the bounds, maximize the conditional objective over c using golden-section search.
- Evaluate off-grid a' by interpolating the next-age expected value.

```
ch(:,t) = solve_golden('wfunc', cminh, cmaxh, ...
                       swh, ind2h, EVh, p, 'h');
wh(:,t) = wfunc(ch(:,t), swh, ind2h, EVh, p, 'h');
```

The search produces both optimal consumption and the value of that decision.

The code searches over consumption; liquid saving follows from the budget.

Housing Choices Reuse the Conditional Values

Action in `solveh.m`	Numerical calculation
1. Rent	Evaluate renter value at resources after any house sale
2. Buy without debt	Search over 7 house sizes
3. Buy with debt	Search over house sizes and 5 original mortgage ratios
4. Refinance	Retain the home and search over mortgage ratios
5. Stay inactive	Compare feasible repayment choices and the exact minimum payment

- Each candidate determines L and the next mortgage and housing position.
- Evaluate its optimized W_t and retain the best candidate within the action.
- Existing renters evaluate actions 1–3. Owners evaluate all five.

Mortgage Constraints in the Numerical Search

For a new loan at $p = 1$, financed balance is $\theta' h'$. Proceeds are

$$\boxed{b^{\text{new}} = \frac{\theta' h' - f_{0,m}}{1 + f_{1,m}}.}$$

- Test LTV and PTI on proceeds: $b^{\text{new}} \leq \theta_m h'$ and $\kappa_m b^{\text{new}} \leq \theta_y(t) y$.
- A new loan sets $\omega' = 1$ and payment $\bar{m}' = \kappa_m \theta' h'$.
- Without refinancing, retain θ, h and compare remaining-balance choices ω'.

$$\omega'_{\text{minimum payment}} = \max\{(1 + r_m)\omega - \kappa_m, 0\}.$$

The code evaluates this exact boundary even when it lies between grid nodes.

▸ Boundary interpolation

Integrating the Mortgage Utility Cost

Let the best values before the cost draw be

$$V_N = \max\{V^1, V^2, V^5\}, \qquad V_M = \max\{V^3, V^4\}.$$

In the code, $\xi \sim \text{Exponential}(\lambda_\xi)$ with $\lambda_\xi = \texttt{p.nu} = 3$.

$$P_M = 1 - \exp[-\lambda_\xi \max\{V_M - V_N, 0\}].$$

The expected decision-stage value is

$$V_t = \begin{cases} V_N, & V_M < V_N, \\ V_M - P_M/\lambda_\xi, & V_M \geq V_N. \end{cases}$$

- Origination occurs when its value gain exceeds the utility cost.
- The paper's mean-cost convention gives $\nu = 1/\lambda_\xi = 1/3$ here.
- The code assigns probabilities to the best action in each group.

Expectations Complete the Age-t Solution

After comparing actions and integrating ξ, the code averages over the remaining shocks.

- **Maintenance:** evaluate owner values at $\widetilde{a}_j = (1 + r_l(a))a - \delta_j h$, then average with $\pi_{\delta,j}$.
- **Transitory income:** average across the 3 income nodes with quadrature weights w_e.
- **Persistent income:** multiply by the transition probabilities in F_{zz}.

$$\bar{V}_t(\text{portfolio}, z_{\text{prev}}) = \sum_{z,e,\delta} F_{zz}(z_{\text{prev}}, z) w_e \pi_\delta V_t(\text{adjusted resources}, \text{portfolio}, z, e).$$

This expected value becomes the continuation object for the age-$(t-1)$ problem.

Where Interpolation Enters

Evaluation	Why it can fall between nodes
Continuation value at a'	Golden-section search chooses continuous consumption and implied saving
Conditional value at L	A housing or mortgage action changes available liquidity
Owner value at $\widetilde{a}$	Returns and maintenance shift liquid resources
Minimum-payment value	The exact remaining mortgage fraction can be off-grid

The baseline uses linear interpolation. Some interpolants also allow linear extrapolation. Grid bounds and resolution affect accuracy, so feasibility and boundary checks matter.

① Numerical Representation

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From Solved Values to Household Histories

- `simulate.m` sets $N = 25{,}000$ and uses $2N = 50{,}000$ household histories with paired shock draws.
- Households initially rent and hold zero liquid wealth. Initial persistent income follows its ergodic distribution.
- Move forward over all 244 quarterly ages, drawing income and maintenance shocks.
- At each state, call `solveh` using the age-specific interpolated W_t values.
- Draw the discrete action from its probabilities, then interpolate conditional consumption and update the portfolio.

The simulation re-evaluates housing choices at realized states rather than simply reading a stored action index.

One Household's Forward Update

For household i at age t:

1. Form adjusted resources: $\widetilde{a}_{it} = (1 + r_l(a_{it}))a_{it} - \delta_{it}h_{it}$.
2. Evaluate feasible housing actions and their probabilities at current income.
3. Select an action and obtain L_{it} and $(\omega_{i,t+1}, \theta_{i,t+1}, h_{i,t+1})$.
4. Interpolate consumption, clip it to feasible bounds, and use `savings` to recover $a_{i,t+1}$.

$$\boxed{b_{i,t+1} = \omega_{i,t+1}\theta_{i,t+1}h_{i,t+1}.}$$

The next quarter starts with this updated portfolio and new shocks.

Distribution, Moments, and Experiments

- Household histories generate the asset and debt distributions at each age.
- The baseline simulation reports calibration moments and other balance-sheet statistics.
- Experiment scripts use the solved objects and simulated population to evaluate interventions.

Driver selection	Script
`'figure4'`	`simulate_liquidity.m`
`'figure5'` or `'table7'`	`simulate_taxrebate.m`

The supplied driver selects `'figure5'`.
Forward household simulation generates an empirical distribution rather than updating a full distribution array over the state grid.

What the Algorithm Computes

- **State-space grids** approximate household value and consumption functions.
- **Backward induction** incorporates future risks into each age's optimal decisions.
- **Conditional values** make it possible to reuse consumption–saving solutions across housing actions.
- **Forward simulation** generates portfolio histories and model moments.

Optimal choices at each state first, population outcomes second.

Renter Consumption and Saving

◂ Conditional saving

$$h^r(c) = \left(\frac{\alpha}{R}\right)^{1/\sigma} c, \qquad n(c) = \phi^{1/\gamma} c^{-\sigma/\gamma}.$$

The renter's budget gives

$$\boxed{a' = L + \phi^{1+1/\gamma} c^{-\sigma/\gamma} - \left(1 + \alpha^{1/\sigma} R^{1-1/\sigma}\right) c.}$$

Substitution into utility yields the objective used by `wfunc.m`:

$$W_t^R(L, z) = \max_c \left\{ \left(1 + \alpha^{1/\sigma} R^{1-1/\sigma}\right) \frac{c^{1-\sigma}}{1-\sigma} - \frac{\phi^{1+1/\gamma}}{1+\gamma} c^{-\sigma(1+1/\gamma)} + \beta \bar{V}_{t+1}^R(a', z) \right\}.$$

Interpolation at the Minimum-Payment Boundary

◂ Mortgage constraints

Let $\omega^* = \max\{(1 + r_m)\omega - \kappa_m, 0\}$ fall between ω_j and ω_{j+1}.

$$\lambda = \frac{\omega_{j+1} - \omega^*}{\omega_{j+1} - \omega_j}.$$

At the liquidity L^* implied by that exact payment, evaluate

$$W_t(L^*, \omega^*, \ldots) \approx \lambda W_t(L^*, \omega_j, \ldots) + (1 - \lambda) W_t(L^*, \omega_{j+1}, \ldots).$$

- The code separately tests grid choices that repay more than required.
- It then adds the exact minimum-payment candidate to the comparison.
- This preserves the economically important repayment boundary.

Code Map and Scope

File	Function in the solution
`start.m`	Parameters, grids, terminal condition, backward recursion, driver
`interest.m`	Exogenous return schedule on liquid assets
`savings.m`	Budget mapping from consumption to saving
`wfunc.m`	Conditional consumption–saving objective
`solve_golden.m`	Vectorized golden-section optimization
`solveh.m`	Housing and mortgage alternatives and cost integration
`simulate.m`	Forward histories and model moments
`simulate_taxrebate.m`	Cash-transfer experiment
`simulate_liquidity.m`	Home-equity liquidity experiment

These slides document the supplied `Main nu 3` implementation.
CompEcon supplies numerical helpers for quadrature, bisection, and grid operations. The baseline normalizes the house price to $p = 1$.